# Team Frequency Assignments

## IN-SPACe CAN-7USAT/Model Rocketry India Student Competition 2026

**Your Can7USat /Model Rocket has two telemetry links - an 868 MHz LoRa link and a 2.4 GHz XBee link, each with a MAIN and a BACKUP channel.** You may use either link or both. Use MAIN by default; switch to BACKUP only when Net Control tells you. Never transmit on anything not listed as yours below.

## Strictly avoid - never transmit on these

- Any frequency or channel not listed as yours below (868 main/backup, or your XBee main/backup).
- The 433 MHz band (419 - 437 MHz) - reserved for rocket ignition command
- The 915 MHz band - reserved for rocket ignition command
- Voice / walkie-talkie band: 446.0 - 446.2 MHz.
- 2405 MHz (XBee Ch 11) - reserved for drone operations.
- Transmitting before your slot is called, or leaving any transmitter switched on after you finish.
- All CanSats must automatically switch off transmission 90 min after launch detection, to avoid interference with other teams.

## LoRa settings (868 MHz link)

| Parameter | Value |
|---|---|
| **Modulation** | LoRa (not FSK) |
| **Band** | 865 - 868 MHz (India licence-free) |
| **Bandwidth** | 125 kHz (mandatory - India caps carrier at 200 kHz) |
| **Spreading Factor** | Your value SF7-SF11 (see your row) |
| **Coding Rate** | 4/5 |
| **Sync Word** | 0x80 + Team ID (see your row) |
| **TX Power** | +10 dBm on pad / ~+14 dBm in flight |

*Each pad has a different SF (SF7-SF11). In this narrow band the different spreading factors keep the five live links separable even where channels are close. The Sync Word makes your receiver ignore other teams on the same frequency.*

## XBee settings (2.4 GHz link)

Use the 2.4 GHz XBee variant in 802.15.4 mode (NOT XBee 900). Your Main and Backup channels and your PAN ID are in the table below. The PAN ID (0x1000 + your team ID) is unique to your team. Use your Main channel by default; move to Backup only on Net Control's call.

## Ground station & remote control

- Ground station: use a high-gain directional antenna (e.g., a Yagi), elevated on a mast, for reliable reception at 1 km and beyond. Use one receiver per channel (or an SDR).
- **Remote on/off (telecommand):** your CanSat must be able to switch EACH link (868 LoRa and 2.4 GHz XBee) ON and OFF by telecommand, so you can silence a link if Net Control reports it is interfering with another team.

## Video transmitters (VTX)

- Any onboard video transmitter must NOT operate in any competition or reserved band (433 MHz rockets, 865-868 MHz telemetry, 915 MHz ignition, 2405 MHz drone, the 2.4 GHz XBee channels, or 446 MHz voice). A 5.8 GHz VTX is usually the safest choice.
- **Jury check required:** have your exact video transmitter frequency measured and approved by the jury before you are cleared to fly.

## Your assigned CanSat settings

**Programme your 868 link (Main/Backup + SF), your 2.4 GHz XBee link (Main/Backup channel + PAN ID), and your team Sync Word.**

| Sr.No. | Team Number 2026-INSPACe-CANSAT-XXX | 868 Main (MHz) | 868 Backup (MHz) | SF | Sync | XBee Main (MHz) | XBee Backup (MHz) | PAN ID |
|---|---|---|---|---|---|---|---|---|
| 1. | 001 | 865.50 | 867.25 | SF7 | 0x81 | Ch 17 (2435) | Ch 21 (2455) | 0x1001 |
| 2. | 002 | 866.00 | 867.75 | SF8 | 0x82 | Ch 15 (2425) | Ch 25 (2475) | 0x1002 |
| 3. | 003 | 866.50 | 865.25 | SF9 | 0x83 | Ch 19 (2445) | Ch 14 (2420) | 0x1003 |
| 4. | 004 | 867.00 | 865.75 | SF10 | 0x84 | Ch 23 (2465) | Ch 16 (2430) | 0x1004 |
| 5. | 005 | 867.50 | 866.25 | SF11 | 0x85 | Ch 26 (2480) | Ch 20 (2450) | 0x1005 |
| 6. | 009 | 865.50 | 867.25 | SF7 | 0x89 | Ch 17 (2435) | Ch 21 (2455) | 0x1009 |
| 7. | 010 | 866.00 | 867.75 | SF8 | 0x8A | Ch 15 (2425) | Ch 25 (2475) | 0x100A |
| 8. | 011 | 866.50 | 865.25 | SF9 | 0x8B | Ch 19 (2445) | Ch 14 (2420) | 0x100B |
| 9. | 014 | 867.00 | 865.75 | SF10 | 0x8E | Ch 23 (2465) | Ch 16 (2430) | 0x100E |
| 10. | 016 | 867.50 | 866.25 | SF11 | 0x90 | Ch 26 (2480) | Ch 20 (2450) | 0x1010 |
| 11. | 017 | 865.50 | 867.25 | SF7 | 0x91 | Ch 17 (2435) | Ch 21 (2455) | 0x1011 |
| 12. | 018 | 866.00 | 867.75 | SF8 | 0x92 | Ch 15 (2425) | Ch 25 (2475) | 0x1012 |
| 13. | 020 | 866.50 | 865.25 | SF9 | 0x94 | Ch 19 (2445) | Ch 14 (2420) | 0x1014 |
| 14. | 021 | 867.00 | 865.75 | SF10 | 0x95 | Ch 23 (2465) | Ch 16 (2430) | 0x1015 |
| 15. | 022 | 867.50 | 866.25 | SF11 | 0x96 | Ch 26 (2480) | Ch 20 (2450) | 0x1016 |
| 16. | 023 | 865.50 | 867.25 | SF7 | 0x97 | Ch 17 (2435) | Ch 21 (2455) | 0x1017 |
| 17. | 024 | 866.00 | 867.75 | SF8 | 0x98 | Ch 15 (2425) | Ch 25 (2475) | 0x1018 |
| 18. | 025 | 866.50 | 865.25 | SF9 | 0x99 | Ch 19 (2445) | Ch 14 (2420) | 0x1019 |
| 19. | 027 | 867.00 | 865.75 | SF10 | 0x9B | Ch 23 (2465) | Ch 16 (2430) | 0x101B |
| 20. | 028 | 867.50 | 866.25 | SF11 | 0x9C | Ch 26 (2480) | Ch 20 (2450) | 0x101C |
| 21. | 029 | 865.50 | 867.25 | SF7 | 0x9D | Ch 17 (2435) | Ch 21 (2455) | 0x101D |
| 22. | 030 | 866.00 | 867.75 | SF8 | 0x9E | Ch 15 (2425) | Ch 25 (2475) | 0x101E |
| 23. | 031 | 866.50 | 865.25 | SF9 | 0x9F | Ch 19 (2445) | Ch 14 (2420) | 0x101F |
| 24. | 032 | 867.00 | 865.75 | SF10 | 0xA0 | Ch 23 (2465) | Ch 16 (2430) | 0x1020 |
| 25. | 033 | 867.50 | 866.25 | SF11 | 0xA1 | Ch 26 (2480) | Ch 20 (2450) | 0x1021 |
| 26. | 034 | 865.50 | 867.25 | SF7 | 0xA2 | Ch 17 (2435) | Ch 21 (2455) | 0x1022 |
| 27. | 035 | 866.00 | 867.75 | SF8 | 0xA3 | Ch 15 (2425) | Ch 25 (2475) | 0x1023 |
| 28. | 036 | 866.50 | 865.25 | SF9 | 0xA4 | Ch 19 (2445) | Ch 14 (2420) | 0x1024 |
| 29. | 038 | 867.00 | 865.75 | SF10 | 0xA6 | Ch 23 (2465) | Ch 16 (2430) | 0x1026 |
| 30. | 039 | 867.50 | 866.25 | SF11 | 0xA7 | Ch 26 (2480) | Ch 20 (2450) | 0x1027 |
| 31. | 040 | 865.50 | 867.25 | SF7 | 0xA8 | Ch 17 (2435) | Ch 21 (2455) | 0x1028 |
| 32. | 041 | 866.00 | 867.75 | SF8 | 0xA9 | Ch 15 (2425) | Ch 25 (2475) | 0x1029 |
| 33. | 042 | 866.50 | 865.25 | SF9 | 0xAA | Ch 19 (2445) | Ch 14 (2420) | 0x102A |
| 34. | 044 | 867.00 | 865.75 | SF10 | 0xAC | Ch 23 (2465) | Ch 16 (2430) | 0x102C |
| 35. | 046 | 867.50 | 866.25 | SF11 | 0xAE | Ch 26 (2480) | Ch 20 (2450) | 0x102E |
| 36. | 048 | 865.50 | 867.25 | SF7 | 0xB0 | Ch 17 (2435) | Ch 21 (2455) | 0x1030 |
| 37. | 050 | 866.00 | 867.75 | SF8 | 0xB2 | Ch 15 (2425) | Ch 25 (2475) | 0x1032 |
| 38. | 053 | 866.50 | 865.25 | SF9 | 0xB5 | Ch 19 (2445) | Ch 14 (2420) | 0x1035 |
| 39. | 054 | 867.00 | 865.75 | SF10 | 0xB6 | Ch 23 (2465) | Ch 16 (2430) | 0x1036 |
| 40. | 056 | 867.50 | 866.25 | SF11 | 0xB8 | Ch 26 (2480) | Ch 20 (2450) | 0x1038 |

| Sr.No. | Team Number 2026-INSPACe-CANSAT-XXX | 868 Main (MHz) | 868 Backup (MHz) | SF | Sync | XBee Main (MHz) | XBee Backup (MHz) | PAN ID |
|---|---|---|---|---|---|---|---|---|
| **41.** | **060** | 865.50 | 867.25 | SF7 | 0xBC | Ch 17 (2435) | Ch 21 (2455) | 0x103C |
| **42.** | **062** | 866.00 | 867.75 | SF8 | 0xBE | Ch 15 (2425) | Ch 25 (2475) | 0x103E |
| **43.** | **063** | 866.50 | 865.25 | SF9 | 0xBF | Ch 19 (2445) | Ch 14 (2420) | 0x103F |
| **44.** | **064** | 867.00 | 865.75 | SF10 | 0xC0 | Ch 23 (2465) | Ch 16 (2430) | 0x1040 |
| **45.** | **065** | 867.50 | 866.25 | SF11 | 0xC1 | Ch 26 (2480) | Ch 20 (2450) | 0x1041 |
| **46.** | **066** | 865.50 | 867.25 | SF7 | 0xC2 | Ch 17 (2435) | Ch 21 (2455) | 0x1042 |
| **47.** | **067** | 866.00 | 867.75 | SF8 | 0xC3 | Ch 15 (2425) | Ch 25 (2475) | 0x1043 |
| **48.** | **069** | 866.50 | 865.25 | SF9 | 0xC5 | Ch 19 (2445) | Ch 14 (2420) | 0x1045 |
| **49.** | **037** | 867.00 | 865.75 | SF10 | 0xA7 | Ch 23 (2465) | Ch 16 (2430) | 0x1025 |

*Assignments repeat across teams because the launch heats ensure teams sharing a channel are never live at the same time (only 5 pads live at once). Your pad follows your SF: SF7 = Pad 1, SF8 = Pad 2, SF9 = Pad 3, SF10 = Pad 4, SF11 = Pad 5.*